Exercise on Logic Circuit Design

Christian Rinderknecht

11 October 2006

1 Conversion

Question. Convert from decimal to 2-complement 8-bit binary numbers.

(a)	105	(b)	-28	(c)	76
(d)	-25	(e)	-101	(f)	127

(a) 105

$$\begin{array}{r|l} 105 & 1 \\ 52 & 0 \\ 26 & 0 \\ 13 & 1 \\ 6 & 0 \\ 3 & 1 \\ 1 & 1 \\ 0 & \end{array}$$

$$105_{10} = 01101001_2$$

(b) -28

$$\begin{array}{r|l} 28 & 0 \\ 14 & 0 \\ 7 & 1 \\ 3 & 1 \\ 1 & 1 \end{array}$$

$$\begin{aligned} 28_{10} &= 00011100_2 \\ -28_{10} &= \overline{00011100} + 1 \\ &= 11100011 + 1 \\ &= 11100100_2 \end{aligned}$$

(c) 76

$$\begin{array}{r|l} 76 & 0 \\ 38 & 0 \\ 19 & 1 \\ 9 & 1 \\ 4 & 0 \\ 2 & 0 \\ 1 & 1 \end{array}$$

$$76_{10} = 01001100_2$$

(d) -25

$$\begin{array}{r|l} 25 & 1 \\ 12 & 0 \\ 6 & 0 \\ 3 & 1 \\ 1 & 1 \\ 0 & \end{array}$$

$$\begin{aligned} 25_{10} &= 00011001_2 \\ -25_{10} &= \overline{00011001} + 1 \\ &= 11100110 + 1 \\ &= 11100111_2 \end{aligned}$$

(e) -101

$$\begin{array}{r|l} 101 & 1 \\ 50 & 0 \\ 25 & 1 \\ 12 & 0 \\ 6 & 0 \\ 3 & 1 \\ 1 & 1 \end{array}$$

$$\begin{aligned} 101_{10} &= 01100101_2 \\ -101_{10} &= \overline{01100101} + 1 \\ &= 10011010 + 1 \\ &= 10011011_2 \end{aligned}$$

(f) 127

$$\begin{array}{r|l} 127 & 1 \\ 63 & 1 \\ 31 & 1 \\ 15 & 1 \\ 7 & 1 \\ 3 & 1 \\ 1 & 1 \end{array}$$

$$127_{10} = 01111111_2$$

2 Subtraction of 2-complement binary numbers

Question. Subtract the following 8-bit, 2-complement binary numbers. State if there is an overflow or an underflow.

(a)
$$\begin{array}{r} 1\,0\,0\,1\,0\,0\,1\,1 \\ -\,\underline{0\,1\,1\,0\,1\,1\,0\,1} \end{array}$$

(b)
$$\begin{array}{r} 1\,0\,0\,1\,1\,0\,1\,1 \\ -\,\underline{1\,0\,0\,1\,1\,0\,1\,1} \end{array}$$

(c)
$$\begin{array}{r} 0\,1\,0\,1\,1\,0\,1\,1 \\ -\,\underline{0\,1\,0\,1\,1\,0\,1\,0} \end{array}$$

(d)
$$\begin{array}{r} 1\,0\,0\,1\,1\,0\,1\,1 \\ -\,\underline{1\,1\,0\,1\,1\,0\,0\,1} \end{array}$$

(e)
$$\begin{array}{r} 1\,1\,1\,1\,1\,1\,0\,1 \\ -\,\underline{0\,1\,1\,0\,1\,1\,0\,1} \end{array}$$

(f)
$$\begin{array}{r} 1\,1\,0\,1\,0\,0\,1\,1 \\ -\,\underline{1\,1\,1\,0\,0\,0\,1\,1} \end{array}$$

Answer.

(a)
$$\begin{array}{ccccccccc} & 1&0&0&1&0&0&1&1 \\ - & 0&1&1&0&1&1&0&1 \\ \hline \mathbf{1} & & &1& & &1&1& \\ & 1&0&0&1&0&0&1&1 \\ + & 1&0&0&1&0&0&1&0 \\ + & & & & & & & &1 \\ \hline & 0&0&1&0&0&1&1&0 \end{array}$$
underflow

(b)
$$\begin{array}{ccccccccc} & 1&0&0&1&1&0&1&1 \\ - & 1&0&0&1&1&0&1&1 \\ \hline \mathbf{1} & \mathbf{1}&1&1&1&1&1&1& \\ & 1&0&0&1&1&0&1&1 \\ + & 0&1&1&0&0&1&0&0 \\ + & & & & & & & &1 \\ \hline & 0&0&0&0&0&0&0&0 \end{array}$$
Obviously!

(c)
$$\begin{array}{ccccccccc} & 0&1&0&1&1&0&1&1 \\ - & 0&1&0&1&1&0&1&0 \\ \hline \mathbf{1} & \mathbf{1}&1&1&1&1&1&1& \\ & 0&1&0&1&1&0&1&1 \\ + & 1&0&1&0&0&1&0&1 \\ + & & & & & & & &1 \\ \hline & 0&0&0&0&0&0&0&1 \end{array}$$

(d)
$$\begin{array}{ccccccccc} & 1&0&0&1&1&0&1&1 \\ - & 1&1&0&1&1&0&0&1 \\ \hline & &1&1&1&1&1&1& \\ & 1&0&0&1&1&0&1&1 \\ + & 0&0&1&0&0&1&1&0 \\ + & & & & & & & &1 \\ \hline & 1&1&0&0&0&0&1&0 \end{array}$$

(e)
$$\begin{array}{ccccccccc} & 1&1&1&1&1&1&0&1 \\ - & 0&1&1&0&1&1&0&1 \\ \hline \mathbf{1} & \mathbf{1}&1&1&1&1&1&1& \\ & 1&1&1&1&1&1&0&1 \\ + & 1&0&0&1&0&0&1&0 \\ + & & & & & & & &1 \\ \hline & 1&0&0&1&0&0&0&0 \end{array}$$

(f)
$$\begin{array}{ccccccccc} & 1&1&0&1&0&0&1&1 \\ - & 1&1&1&0&0&0&1&1 \\ \hline & & &1&1&1&1&1& \\ & 1&1&0&1&0&0&1&1 \\ + & 0&0&0&1&1&1&0&0 \\ + & & & & & & & &1 \\ \hline & 1&1&1&1&0&0&0&0 \end{array}$$

3 Subtraction of BCD numbers

Question.

(a)
$$\begin{array}{ccccc} & 0111 & 0101 & 1001 & 0000 \\ - & 0010 & 1001 & 0011 & 0101 \\ \hline \end{array}$$

(b)
$$\begin{array}{ccccc} & 0011 & 0110 & 0001 & 0111 \\ - & 0001 & 0100 & 0110 & 1000 \\ \hline \end{array}$$

(c)
$$\begin{array}{ccccc} & 1001 & 0101 & 1000 & 0111 \\ - & 0011 & 0101 & 0110 & 0010 \\ \hline \end{array}$$

(d)
$$\begin{array}{ccccc} & 0110 & 1000 & 0011 & 0101 \\ - & 0110 & 1000 & 0101 & 1001 \\ \hline \end{array}$$

(a)

$$
\begin{array}{rcccc}
 & 0 & & 0110 & 111 \\
 & 0111 & 0101 & 1001 & 0000 \\
- & 0010 & 1001 & 0011 & 0101 \\
\hline
 & & 00 & & 0 \\
 & 0100 & 1100 & 0101 & 1011 \\
- & & 0110 & & 0110 \\
\hline
 & 0100 & 0110 & 0101 & 0101
\end{array}
$$

(b)

$$
\begin{array}{rcccc}
 & & 01 & 110 & \\
 & 0011 & 0110 & 0001 & 0111 \\
- & 0001 & 0100 & 0110 & 1000 \\
\hline
 & & & 0 & \\
 & 0010 & 0001 & 1010 & 1111 \\
- & & & 0110 & 0110 \\
\hline
 & 0010 & 0001 & 0100 & 1001
\end{array}
$$

(c)

$$
\begin{array}{rcccc}
 & 01 & & 01 & \\
 & 1001 & 0101 & 1000 & 0111 \\
- & 0011 & 0101 & 0110 & 0010 \\
\hline
 & 0110 & 0000 & 0010 & 0101
\end{array}
$$

(d)

$$
\begin{array}{rcccc}
 & 0110 & 1000 & 0011 & 0101 \\
- & 0110 & 1000 & 0101 & 1001 \\
\hline
 & \multicolumn{4}{c}{\textbf{underflow}}
\end{array}
$$